

Spam Detection: A Machine Learning Approach

Analysis of the UCI Spambase
Dataset

The Project Goal

The Problem: Binary Classification

This project is formulated as a binary classification problem. The primary goal is to train a model that effectively maps an email's feature vector (word frequencies, etc.) to a predicted class label.

The Labels: Spam vs. Non-Spam

Our model will learn to distinguish between two simple, clear labels:

- **Label 0:** Non-Spam (ham)
- **Label 1:** Spam

Why Spam Detection?



A Practical Problem

Spam detection is a real-world, meaningful challenge. The dataset is well-known and suitable for applying and comparing different ML algorithms.



Real-World Impact

An effective classifier improves email security, reduces user exposure to harmful content, and protects against phishing or malware.



Streamlined Management

This project demonstrates how data-driven methods can streamline inbox management and solve everyday challenges in an impactful way.

The UCI Spambase Dataset

4,601

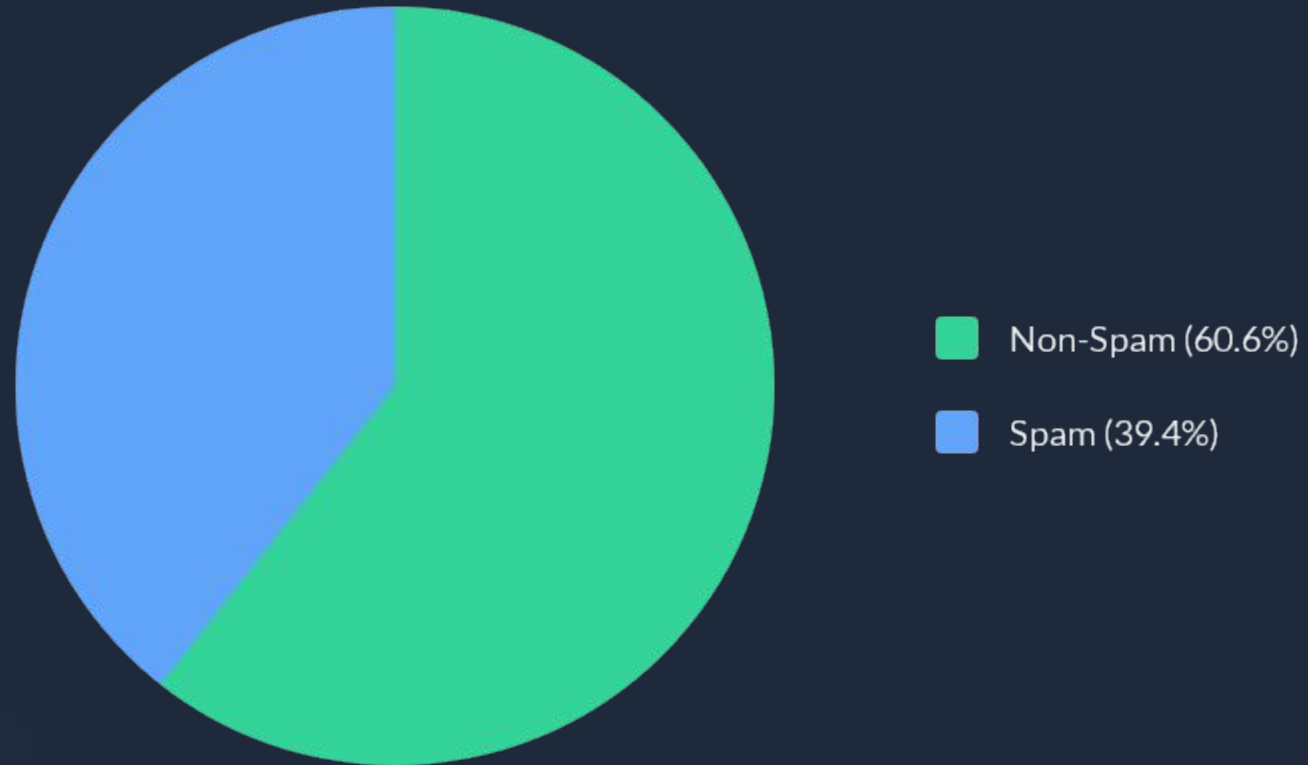
Total Email Samples

58 Total Attributes

This project uses the classic UCI Spambase dataset.

Each sample contains **57 numeric features** (representing word and character frequencies) and **1 target label** (spam or not spam).

Class Distribution



A moderately imbalanced 60/40 split. This is managed using a stratified train-test split to ensure representative samples.

Data Preprocessing Pipeline

-  **Min-Max Scaling:** All 57 numeric attributes were scaled to a $[0, 1]$ range to normalize feature importance.
-  **Data Integrity:** No missing values were found in the dataset, so no feature drops or imputation were required.
-  **Stratified Split:** A strict 80/20 train-test split was performed, preserving the 60/40 class ratio in both sets.
-  **Feature/Label Separation:** Data was separated into features (X) and the target label (y) for model training.

Four Models Were Evaluated



Naive Bayes

A fast, probabilistic baseline model that provides a simple benchmark.

Logistic Regression

A linear, interpretable, and efficient model for binary classification.



Decision Tree

A non-linear model used to capture complex splitting rules in the data.



Random Forest

An ensemble of trees designed for high accuracy and to prevent overfitting.

Model Performance Comparison (Accuracy)



Random Forest significantly outperformed all other models, achieving the highest accuracy on the test set.

Best Model Deep Dive: Random Forest



Accuracy: ~0.96

The model correctly classifies approximately 96% of all emails in the test set.

ROC AUC: 0.986

Excellent class separation ability, indicating a highly robust and reliable model.



Strong Precision/Recall

Effectively minimizes both false positives (good mail as spam) and false negatives (spam as good mail).

Discussion & Conclusion

Conclusion

The **Random Forest** classifier is the optimal model, providing a high-accuracy (95.6%) and robust (0.986 AUC) solution for this spam detection task.

Discussion

Ensemble methods (like RF) effectively captured the non-linear feature interactions. The model's low misclassification rate makes it a reliable choice for a real-world filter.

References

1. UCI Spambase Dataset. (n.d.). UCI Machine Learning Repository.
2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research.

Questions?

Thank you for your time.